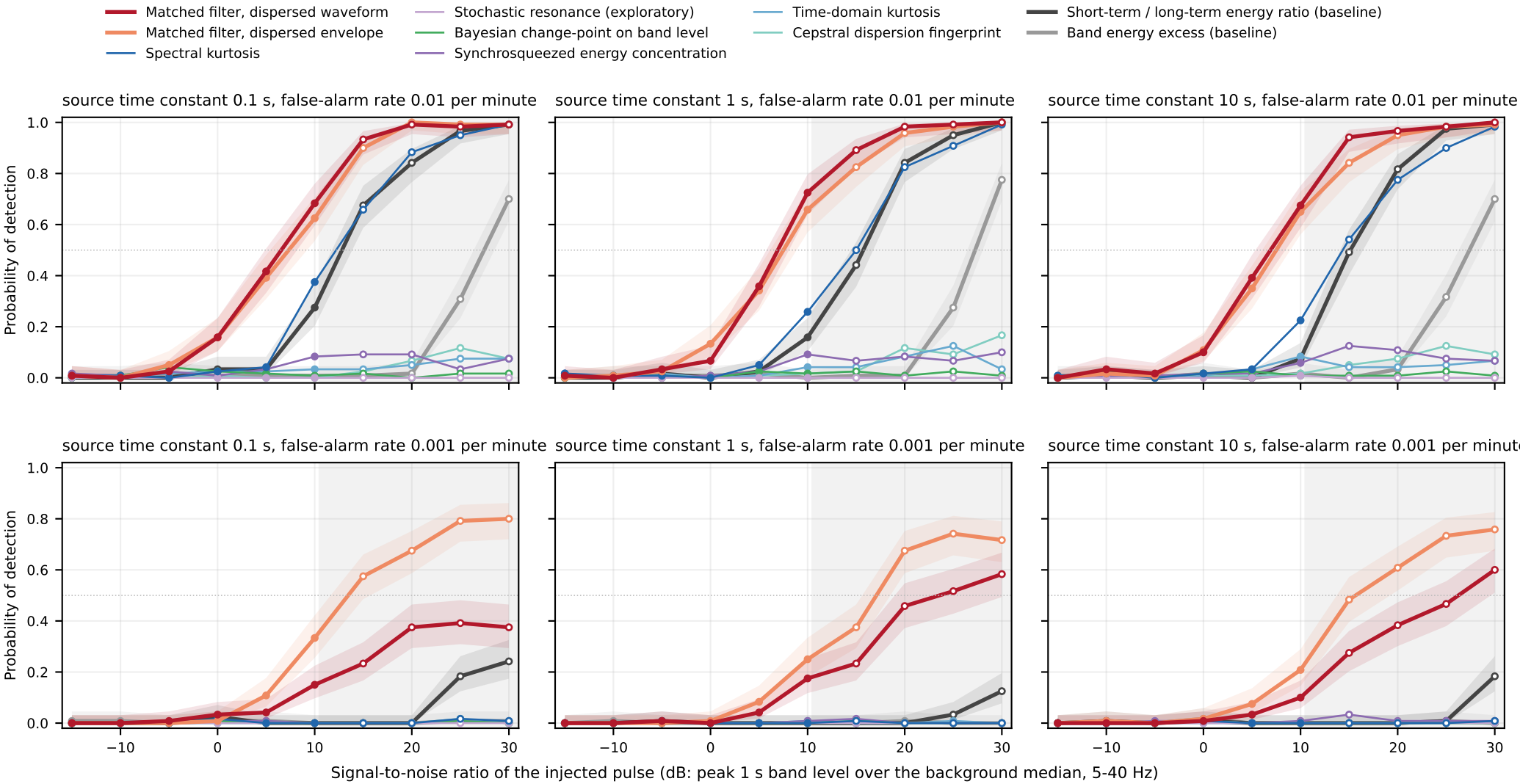
Matched filters built on the dispersed arrival shape detect injected impact pulses 6-9 dB fainter than the energy detectors
at the same false-alarm rate (Perth Canyon logger 3376; the test pulses share the filters' propagation model, so this is their best case)




(i) Synthetic impact sounds were added to 14 days of real recordings from the Perth Canyon sea-noise logger, at six to ten loudness levels, 120 trials each. Each line is a detection method; the shaded grey area shows loudness levels added after the first run (an amendment). Filled circles: pre-registered levels; open circles: added levels. Shaded bands: 95 % Wilson intervals (matched filters and baselines only). Dotted line: 50 % detection. The two matched filters use the expected shape of a long-range deep-ocean arrival; the test pulses were made with the same propagation model, so this is a best case for them.

(ii) Prereg 78c7e88 (test set a), amendment 1 (15-30 dB). Background: IMOS 3376, 1-7 and 9-15 Mar 2014, 5376 non-overlapping 60 s null windows; thresholds = null quantiles at alpha 0.01 / 0.001 per window (0.001 rests on the top ~5 values). Pulse = one-pole source (ESD flat to 1/tau, f^-2 above) x KRAKEN coherent broadband channel (WOA23 March midpoint profile, flat 4,610 m hard bottom, 2-48 Hz, ranges 1,800-3,000 km), 5-40 Hz, 100 Hz. Source posterior: run J seed-1 weighted-median impact for the path geometry only; no 00:19 option, p or eta enters (SNR is the control variable). Caveats C1 (TL slope misfit) and C2 (coupling uncalibrated) apply to any mapping of SNR to impact energy; none is made here.